

Lab 1 – Requirements Engineering: Digital Art Commission & Watermarking Portal

Problem Statement #58 | PES University – Dept. of CSE | SRN: PES1UG24CS916

Actors: Client Buyer, Digital Artist

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall apply a dynamic diagonal overlay watermark to all draft illustrations uploaded by artists before displaying to the client.	High	Pass: Watermarked draft rendered in client portal. Fail: Unwatermarked high-res image exposed before final payment.	Protects artist IP until full payment is received.
FR-002	Functional	The system shall allow a Client Buyer to submit a visual creative brief by filling a structured form with fields for style, resolution, deadline, and reference images.	High	Pass: Brief is saved and artist notified within 60 s. Fail: Brief is lost or artist receives no notification.	Core intake flow — without a brief, the commission cannot begin.
FR-003	Functional	The system shall allow a Digital Artist to upload watermarked WIP progress drafts at each milestone, visible only to the associated Client Buyer.	High	Pass: WIP draft visible to the correct client and invisible to all others. Fail: Draft exposed to unauthenticated users or wrong client.	Enables iterative client feedback while maintaining access control.
FR-004	Functional	The system shall process milestone payments via integrated payment gateway and release the corresponding high-resolution source file download link only upon successful payment confirmation.	High	Pass: High-res file link generated after payment success event. Fail: File accessible before payment or link not generated after payment.	Enforces the pay-to-unlock business model protecting artist revenue.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall allow a Client Buyer to submit revision requests on a WIP draft, with a free revision counter enforced per commission contract.	Medium	Pass: Revision request logged; counter decrements; artist notified. Fail: Revision submitted with no notification or counter exceeding the agreed limit without blocking.	Manages revision scope, preventing unlimited unpaid rework.
NFR-001	Non-Functional (Performance & Security)	Final high-resolution asset downloads must generate expiring signed S3 URLs valid for 60 minutes only.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: URLs remain valid beyond 60 minutes or are accessible without authentication.	Prevents link sharing and unauthorized redistribution of paid assets.
NFR-002	Non-Functional (Availability)	The platform shall maintain 99.5% uptime during peak hours (08:00–22:00 IST) with automated failover, ensuring artists can upload and clients can review without service interruption.	Medium	Pass: Monthly uptime report $\geq 99.5\%$; failover completes in < 30 s. Fail: Downtime exceeds 0.5% in any calendar month or failover > 30 s.	Commissions are time-sensitive; downtime risks deadline breaches and revenue loss.